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The Effects of Corpus Use on Second Language Vocabulary Learning: A Multilevel Meta-analysis

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Abstract

This study investigates the effects of corpus use on second language (L2) vocabulary learning as well as the influence of moderators on effectiveness. Based on 29 studies representing 38 unique samples, all of which met several criteria for inclusion (e.g. with control groups), we found an overall positive medium-sized effect of corpus use on L2 vocabulary learning for both short-term (77 posttest effect sizes; Hedges' $g = 0.74$, $SE = 0.09$, $p < .001$) and long-term periods (34 follow-up effect sizes; Hedges' $g = 0.64$, $SE = 0.17$, $p < .001$). Furthermore, large variation in adjusted mean effect sizes across moderators was revealed. Above all, for the different dimensions of L2 vocabulary knowledge, in-depth knowledge (i.e. referential meanings as well as syntactic features of vocabulary) was associated with a large effect size. Moreover, the results revealed that learners' L2 proficiency and several features of corpus use (i.e. interaction types, corpus types, training, and duration) influence the magnitude of the effectiveness of corpus use in improving L2 vocabulary learning.

INTRODUCTION

Large collections of structured language data (Sinclair 2004), also known as corpora (singular: *corpus*), have been widely used in the field of second language (L2) learning (Biber *et al.* 1998; Johns 1991; Sinclair 1991, 2004) and have served several pedagogical purposes and applications, such as for creating learning materials, designing hands-on activities, and building textbooks, workbooks, and dictionaries (see Flowerdew 1993). In particular, corpus use has been applied to instruction in L2 learning of grammatical and lexical items by providing multiple contextual examples of target items. With such examples, a learner is induced to discover linguistic features by exploring the language data that provide authentic linguistic information necessary for learning (Johns 1991), a process called data-driven

learning (DDL; Johns 1991). More recently, corpus tools (e.g. concordance software) equipped with useful pedagogical features have become widely available, offering students opportunities for hands-on analysis of corpora in classrooms (Godwin-Jones 2001; Reppen 2010).

Though corpus use has many purposes in instructed L2 learning, there has been a particular pedagogical and research focus on its effects on vocabulary learning (see Supplementary material Appendix 1 for a list of L2 vocabulary studies related to corpus use), by drawing on the notion of DDL. These corpus-based vocabulary studies have covered a wide range of methodologies (e.g. characteristics of a given task, pedagogical contexts, and types of measurement) and topics (e.g. suitability of language data, mastery of corpus consultation skills; see the following sections for a detailed review). A synthesis of this rich and varied research can thus inform pedagogical guidelines for the corpus-based approach to L2 vocabulary teaching and learning.

There have been several recent meta-analyses (Cobb and Boulton 2015; Mizumoto and Chujo 2015; Boulton and Cobb 2017) reporting that corpus use had overall positive effects on L2 learning. To further this synthesizing work, in the present study, we conducted a meta-analysis to not only estimate the overall effect of corpus use on L2 vocabulary learning but also find out how much moderating variables influence the estimated effect of corpus use. Our study differs from previous meta-analyses on the effect of corpus use, such that in our study we (i) use the multilevel effect size (ES) calculation (i.e. multiple ESs for each study; ESs nested in studies), (ii) specifically focus on sub-dimensions of L2 vocabulary knowledge, and (iii) use regression analyses not only to statistically compare and contrast mean ESs across moderators (i.e. elements of population, publication, and treatment data of the collected studies) but also to provide adjusted mean ESs for each value of the moderators (Woltman *et al.* 2012; see the 'Data analysis plan' section for details). In the following sections, we review the theoretical background of corpus use as well as previous studies on corpus-based L2 vocabulary learning. Next, we describe the statistical procedures for our multilevel meta-analysis, including the literature search, inclusion criteria, coding procedure, multilevel ES calculation, and data analysis plan. Then, we present the results of the meta-analysis, along with an in-depth discussion of the findings. Last, we discuss the limitations of our study and the implications of our findings.

BACKGROUND

Theoretical background for corpus use in L2 vocabulary learning

The core of corpus use lies in its encouraging learners to construct their L2 knowledge independently by exploring the linguistic data compiled from corpora, such as concordances that provide multiple sentential examples of how a target linguistic item is used (Johns 1994). The inductive process in this definition, then, must involve complex cognitive engagement by learners, including inferencing and hypothesizing on language items (Flowerdew 2015). Given that the process 'presumes that learners build knowledge actively, largely through inductive processes' (Collentine 2000: 45), constructivism can provide theoretical support for corpus use (Cobb 1999). In principle, the constructivist approach rejects the idea of transferring knowledge to students (Collentine 2000), but embraces a notion that learners actively participate in learning, reflecting a paradigm shift in education from teacher-centered to student-centered learning (Kaufman 2004).


council agreed present day value of the award in that case would be in the vicinity of one hundred and twenty thousand pounds. I conclude (pause) that taking
, (SP:JNEPSUNK) If (pause) (------) was going to be in that room (pause) in the vicinity of the bed (pause) presumably two places he could be, under the duvet or
. In British Transport Police may act as constables in, on and in the vicinity of Rail Track's property and my Right Honourable Friend, the Secretary of State
control and jurisdiction over all the the railway system in and on and in the vicinity of the, the railway the railway organisation. In other words they are respons
words and er whether or not the the words er er er er in the vicinity of is more constricting than we anticipate and it is that point which will be
Lord that erm er they w-- w-- will have full jurisdiction on and in the vicinity of the stations which is the point which she which she is anxious about and
travels Walton had noticed that large retail chains tended to locate their stores in the vicinity of large population centres, in the belief that small-town outlets c
, Allen Ginsberg and Gregory Corso briefly congregated there in the Fifties, in the vicinity of the great Paul Bowles. Now, Brion Gysin, Burroughs's fellow denize
lorries,' he said.' Should you happen to be driving in the vicinity of the lorries, extreme caution is advised. A number of motorists have found
smoke? Perhaps Mr Lloyd expects a non-smoker to abstain from breathing while in the vicinity of tobacco smoke. Mr Lloyd's comments are also a slur against t
tackled the Welsh, admittedly naive enough to try to use backrow moves in the vicinity of the French line to a standstill. The hardworking Richard Webster and
's market value, definitely enhanced by his showing against Germany, is in the vicinity of 3 million. That sum was the Bundesliga transfer record as recently as l
of thefts took place when bets were being placed, winnings collected or in the vicinity of public conveniences.' # Robbery ordeal for man # A 29-YEAR-OLD Mal
Lee Clegg (22) told the court he had fired three times in the vicinity of the car driver's head. He did so' to incapacitate him --
two shop stewards. The second order relates to the banning of meetings in the vicinity of the factory gates. The appeal court decided that the case should be h
from these' sealed lorries.' Should you happen to be driving in the vicinity of one of these' sealed lorries' extreme caution is advised as a number
alleged breach of bail conditions. Yet on both occasions he had been in the vicinity of his own pit; unsurprisingly because his house was only fifty yards from it
representations on behalf of local residents unless it actually owns or occupies land in the vicinity of the proposed operating centre. Whereas an objector has a
knows or has reasonable grounds to believe that the (non-visitor) is in the vicinity of the danger concerned or that he may come into the vicinity of the danger
. If the suspect is under arrest, he may be interrogated only in the vicinity of the scene of the crime and in proximity of time to its commission.
the result of a communication from her and shortly thereafter arrested Mr. Bell in the vicinity of Miss Coffey's house. Mr. Bell was taken to a police station in
is no unlawful act. It is because a consideration of the neighbourhood in the vicinity of the highway will lead to the conclusion that the noise or fumes are not
Oman has also complained about the presence of Soviet naval vessels and aircraft in the vicinity of the Straits of Hormuz. The Soviet plan was received more fa
) he knows or has reasonable grounds to believe that the other is in the vicinity of the danger concerned or that he may come into the vicinity of the danger

Figure 1: *A snapshot of concordance lines from the BYU-BNC (Davies 2004). Shown are concordance lines (sample sentences) of 'in the vicinity of'. These sentences are sorted and aligned by 'in the vicinity of'. In addition, the target expression is highlighted and is thus visually salient*

The other unique feature of corpus use is 'Key Word In Context' (KWIC), which is the most common way to use corpora as learning materials. By means of the KWIC function, which is installed in most corpus-based computer programs (e.g. a concordancer), a student can search

for a target linguistic item by typing it, and the programs will display multiple concordance lines of the typed item. For example, Figure 1 is a screenshot of concordance lines of an expression, 'in the vicinity of', excerpted from the British National Corpus presented by Brigham Young University (BYU-BNC; Davies 2004). As the concordance lines expose a learner to multiple incidences of target language items (i.e. frequency effect: Ellis 2002), and the target item is highlighted, the KWIC format, shown in Figure 1, does not only make the concordance lines salient learning input (i.e. input enhancement: Chapelle 2003; Wong 2005) but also increases the possibility that learners will notice target items and be able to acquire them (i.e. noticing hypothesis: Schmidt 2001; Lai and Zhao 2005). Engaging with a large compilation of language data inductively also requires great learner involvement, which makes learning target lexical items easier according to Laufer and Hulstjin's (2001) involvement load hypothesis. Taken together, corpus use can be theoretically supported by constructivism, the inductive approach, and several major second language acquisition frameworks.

Empirical evidence and limitations of corpus use in L2 vocabulary learning

The abovementioned theoretical frameworks have attracted the attention of the empirical researchers who are investigating the effectiveness of corpus use for improving different dimensions of L2 vocabulary knowledge. For example, Cobb (1997, 1999) conducted an empirical study with two different vocabulary learning conditions: (i) using corpus tools for vocabulary learning (experimental condition) and (ii) using dictionaries and word lists (control condition—traditional vocabulary learning). The results of Cobb's study with adult Omani L2 learners showed that using corpus tools yielded more gains in terms of the learners' L2 vocabulary knowledge, such as their definitional knowledge and their ability to transfer their word knowledge to other contexts. In a study by Chan and Liou (2005), adult Taiwanese L2 learners completed Web-based practice units using a bilingual corpus tool, and the results showed significant improvement in the participants' knowledge of collocation. In addition, Frankenberg-Garcia's studies (2012, 2014) confirmed the positive effects of using concordance lines excerpted from corpus data on L2 vocabulary learning. The results indicated that these examples were effective for L2 learners in terms of understanding target vocabulary (2012, 2014), correcting typical L2 mistakes (2012), and writing sentences using target vocabulary (2014).

However, corpus use for L2 vocabulary learning is not without limitations. Previously, Boulton (2010) suggested *barrier* as an umbrella term for the limitations of corpus use in language learning, including (i) new material (e.g. KWIC format), (ii) technology (e.g. concordancer), and (iii) learning approaches (e.g. inductive learning), all of which appear to be germane to the suitability of language data and corpus consultation skills. For example, it

has been previously suggested that learners' linguistic inferences from contexts (e.g. concordance lines) could be fallacious (Schmitt 2008) and that sometimes learners retain these inaccurate inferences in their lexicons (Mondria 2003). Thus, concordance lines given in corpus use should be comprehensible to learners and should provide enough contextual clues for learners' processing of target lexical items for their linguistic inquiries. In other words, they must be suitable for target learners. Moreover, learners might need to understand how to effectively use concordance lines or corpus tools for their L2 vocabulary learning practice.

Considerations about corpus use in L2 vocabulary learning

When the suitability of language data is evaluated, one may observe that a list of concordance lines automatically retrieved from corpus data has an arbitrary proportion of sentences that give strong, relevant, or useful information about target vocabulary (i.e. high-constraint sentences; Adlof *et al.* 2016) and/or sentences that are comprehensible to learners. For this reason, learners may be cognitively burdened if the language in the sentences is either low-constraint or beyond the learners' current L2 proficiency level (Allan 2009; Lee *et al.* 2017).

With this in mind, several researchers in the field of L2 vocabulary learning have emphasized the importance of carefully selecting corpus data for teaching L2 vocabulary. For example, Frankenberg-Garcia (2012, 2014) and Lee *et al.* (2017) identified sample sentences that contain enough contextual clues about the target lexical items and confirmed that the use of such materials resulted in positive learning outcomes. Allan (2009) recommended that teachers use corpus-informed materials in the form of graded readers, which are more finely tuned to learners' L2 proficiency levels, rather than using pre-established reference corpora, such as the Corpus of Contemporary American English (COCA), BNC, Brown Corpus, and Open American National Corpus (OANC). On similar grounds, Lee *et al.* (2015) developed a corpus tool designed to assist teachers to easily upload language data to build suitable corpus-informed materials for their students. Taken together, the suitability of language data in terms of adapting corpus use for L2 vocabulary learning depends on the learners' L2 proficiency levels, the type of concordance lines (i.e. carefully selected for comprehension or automatically retrieved), and type of learning material (i.e. corpus-informed materials or reference corpora).

To evaluate the learners' mastery of corpus consultation skills, it has been argued that the KWIC format can hinder effective implementation of corpus use for learners (see Sinclair 2003) because the presentation of 'unfinished sentences' (Johns 1986: 157) in this format (see Figure 1) could be unfamiliar to learners. Indeed, concordance lines are not designed for traditional text-reading strategy (e.g. linear reading; Boulton and Cobb 2017), so providing learners with a clear guideline about how to explore language data would be helpful (Gavioli

2005; Boulton 2009; Lee *et al.* 2017). For this reason, some previous studies attempted to help learners handle corpus materials or activities by manipulating instructional or methodological features. For example, Boulton (2010) examined the positive effect of using paper-based materials in DDL on L2 vocabulary learning. By using prepared concordance printouts, Boulton aimed to alleviate learners' burden of dealing with concordance lines in rather unfamiliar computer-assisted environments. Similarly, Lee *et al.* (2017) implemented a training workshop that aimed to help students better understand how to use concordance lines for learning L2 vocabulary. However, Boulton (2009) found that the form of KWICs could be effective, even for untrained learners. These apparently contradictory findings about corpus use in language learning call for a meta-analysis that can statistically combine, compare, and contrast the effects of corpus use on language learning in terms of their different methodologies.

Rationale of the present study

Recent meta-analyses have pointed to the value of corpus use in L2 learning (Cobb and Boulton 2015; Mizumoto and Chujo 2015; Boulton and Cobb 2017). For example, Cobb and Boulton's (2015) preliminary meta-analysis included 21 corpus-use studies. They found the overall average ES to be large (Cohen's d = 1.04 for between-group ES; d = 1.68 for within-group ES).[1] In contrast, Mizumoto and Chujo's (2015) meta-analysis on the DDL approach to vocabulary learning, based on 14 studies in Japanese EFL contexts, found a small ES (d = 0.90 for within-group ES).

These two studies, however, have their own limitations. First, as Cobb and Boulton (2015) acknowledged, their meta-analysis was preliminary, in that they calculated only the overall average ES, without taking into account the different methodologies used in the included studies. Mizumoto and Chujo (2015) based their analyses on just one population (Japanese EFL learners), so their conclusion on the effect of DDL on L2 vocabulary learning cannot be generalized to other populations. Furthermore, their meta-analysis did not respond to the call from Cobb and Boulton (2015) about the need to perform moderator estimation.

As a more complete version of Cobb and Boulton (2015), Boulton and Cobb's (2017) meta-analysis included 64 studies representing 88 unique samples and found large ESs for both between-group (d = 0.95) and within-group comparisons (d = 1.50). Furthermore, they analyzed moderator variables and found that corpus use had positive effects in almost any language learning situation. However, as in the two previous meta-analyses, Boulton and Cobb selected only a single ES per study for the data analysis, a standard meta-analysis approach in the field, and thus could explore methodological differences only across the included studies.

Our study extends the previous meta-analyses to investigate the positive effect of corpus use on L2 learning, but differs from them in our narrower focus on L2 vocabulary learning. Specifically, we explored how corpus use could improve the different dimensions of L2 vocabulary knowledge as part of the analysis of moderator variables. To this end, we chose Henriksen's framework (1999) to distinguish L2 vocabulary knowledge across the following three major dimensions: (i) precise knowledge, that is, being able to understand the definition of a target lexical item, (ii) in-depth knowledge, that is, being able to state the referential meaning (e.g. synonyms, antonyms) as well as syntactic features (e.g. collocational patterns) of a target lexical item, and (iii) productive use ability, that is, being able to construct a sentence using a target lexical item. Among several other perspectives on L2 vocabulary knowledge, we chose Henriksen's framework because it is widely used; most L2 vocabulary assessments are based on this framework (Gyllstad 2013).

Furthermore, our study is unique in the methodology of meta-analysis used. Instead of computing a single ES per study, we carried out a multilevel meta-analysis (i.e. multiple ESs per study; two-level model) to fully capture large methodological differences *within* (i.e. between effect sizes within studies) and across the included studies. This approach, of course, may raise an issue of statistical independence between the multiple ESs of a study; however, using a multilevel regression, analysis should resolve the issue (see Hox *et al.* 2010; Van den Noortgate *et al.* 2013; Pratt *et al.* 2016). Also, regression analyses allowed us to statistically compare the different impacts of moderators and to compute adjusted means for each moderator, holding other moderators constant (see the 'Data analysis plan' section for details). In this way, we expect our meta-analysis to provide informative pedagogical implications for the effectiveness of corpus use for L2 vocabulary learning. The following two research questions guided this study:

1 How effective is corpus use in improving L2 vocabulary learning?

2 What are the moderators that influence the magnitude of the effectiveness of corpus use?

METHODS

In an effort to answer the aforementioned research questions, we focused exclusively on studies based on experimental designs with control conditions for corpus use, unlike the previous meta-analysis studies (Cobb and Boulton 2015; Mizumoto and Chujo 2015; Boulton and Cobb 2017) which included both within-group (i.e. differences among participants who are in the same group; pre/posttest designs with no control groups) and between-group comparisons (i.e. differences between two or more groups of participants; pre/posttest designs with control groups) in their ES calculations. In some instances, only including one data set

(i.e. between-group comparisons) could be less comprehensive than a study based on two (i.e. between-group and within-group comparisons); however, we believe that our decision is more appropriate in our situation where the primary goal is to examine the overall *treatment effect* of corpus use in L2 vocabulary learning (see Shadish *et al.* 2002). To this end, we took multiple steps to collect and evaluate existing studies for inclusion in our meta-analysis, as described below.

Literature search

As the first step, we conducted keyword searches (either corpus, corpora, concordance, or data-driven learning) of the literature written in English available up to the year 2016 in major databases, including Education Resources Information Center, Linguistics and Language Behavior Abstracts, Web of Science, and Dissertation Abstracts Databases. In addition, we manually searched not only in relevant academic journals, including *CALICO Journal*, *Computer-Assisted Language Learning*, *Computers & Education*, *Language Learning & Technology*, *ReCALL*, and *SYSTEM* but also through the reference lists in the collected studies and previous meta-analyses to find unidentified studies. We identified 52 studies after manually excluding studies unrelated to L2 vocabulary learning. Supplementary material Appendix 1 describes the identified 52 studies and states the reasons for excluding others in our meta-analysis.

Inclusion criteria

First, we decided that a study should implement random control trials along with a control group. If this condition is not met, the study must at least investigate and confirm homogeneity between treatment and control groups and could thus be considered to be a quasi-experimental study. We found that over 30 per cent (19 studies) of the 52 studies did not meet the first criterion. In particular, two of them were small case studies, and one study did not control for baseline differences in a quasi-experimental design. Moreover, 16 of the studies did not have control groups. Only 33 studies remained. Second, a study should report descriptive and/or inferential statistics of posttest scores, which are necessary for calculating posttest ESs. Only two studies did not meet the second criterion, so 31 studies remained. Third, more specifically regarding control conditions, a study should have a conventional (or standard) instruction control group with which the effects of corpus use on L2 vocabulary learning in the experimental group could be compared. Another two studies were excluded because they had control groups that received no instruction at all. In the end, only 29 studies fully matched our three criteria (see Supplementary material Appendix 1 for a list of the included studies).

Multilevel ES calculation

Each posttest and follow-up (e.g. delayed posttest) ES was calculated according to Hedges' *g* (Hedges and Olkin 1985; see Supplementary material Appendix 2 for the equations of the ES calculation), which provides more conservative calculations than Cohen's *d* does, particularly for small samples ($n < 50$; Hedges and Olkin 1985; Huber 2013). Hedges' *g* is calculated by multiplying the so-called bias correction factor ($J = 1 - [3/\{4 \times df - 1\}]$; Hedges and Olkin 1985) by Cohen's *d*. For this reason, *g* is also known as 'unbiased *d*' (Boulton and Cobb 2017). As mentioned earlier, we computed more than one ES per study when possible, so the constructed data set is a two-level model, as shown in Figure 2.[2] For example, we computed three posttest ESs for Kaur and Hegelheimer (2005; Study #10), and two posttest ESs for Supatranont (2005; Study #11). We found that Kaur and Hegelheimer not only tested students' in-depth vocabulary knowledge by using two different measurements, which were a cloze test and a sentence-building task, but also examined learners' productive use ability by counting the number of correct words used in students' writing. As a result, Kaur and Hegelheimer had three different measurements in their study, for which we generated three posttest ESs. In line with this approach, in Supatranont's (2005) study, there were two different dependent variables, including definition tests for precise knowledge and cloze tests for in-depth knowledge; therefore, we computed two posttest ESs.

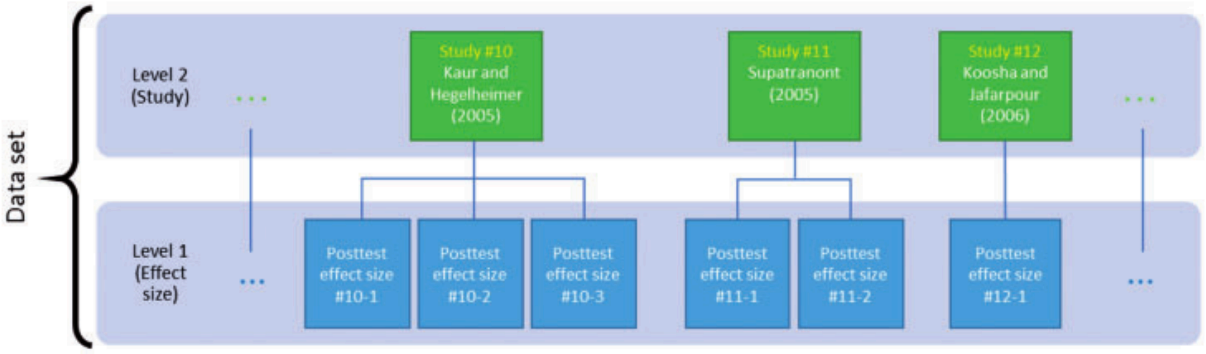


Figure 2: *Structure of data set for multilevel meta-analysis in case of posttest effect sizes*

To check the robustness of our results, we generated gain-score ESs based on pretest and posttest scores whenever a study used a pretest–posttest design instead of a posttest only (Supplementary material Appendix 3). We did so because the pretest–posttest design requires different equations for the ES calculation to include pretest scores in the ES estimation. The posttest differences are often larger than the pretest differences because of treatment effects (i.e. the treatment does not affect everyone the same way; hence there are likely to be larger posttest differences). Using the aforementioned between-group equations that use posttest differences instead of pretest differences would therefore underestimate the ES of pretest–

posttest design studies (see Morris 2008; Plonsky and Oswald 2012). The results of this approach showed that the newly generated gain-score ESs were slightly larger than the posttest ESs, though the difference between them was not statistically significant. Taken together, for the sake of robustness we used gain-score ESs over posttest ESs for the relevant studies.

Publication bias

Given the retrospective nature of a meta-analysis, the inclusion of previous empirical studies is limited to those *identifiable* by a literature search. First, for scholarly journal articles, it has been said that studies with statistically significant findings are more likely to be submitted and accepted for publication and cited than studies with non-significant findings; such a bias might influence the results of meta-analysis. Furthermore, it has been said that small studies are more likely to show larger ESs; therefore, the ESs computed from a pool of small studies are sometimes contradicted by those in later large studies, which are not published as frequently and quickly as small-scale studies are (see Egger *et al.* 1997). For this reason, we assessed any possible publication bias (using STATA 14, see Supplementary material Appendix 4 for details) among the calculated ESs before carrying out full-scale data analyses. Taken together, the results of our checks (i.e. Funnel plot + Egger's test) revealed no significant publication bias among the calculated posttest ESs.

Coding procedure

To build a data set for our meta-analysis, we analyzed the previous meta-analyses to identify and develop a coding rubric for ES calculations and analyses of moderator variables. To this end, we decided to adapt the categories commonly used by other meta-analyses (e.g. publication, population, and treatment data) and to especially endorse Boulton and Cobb's (2017) coding strategy for analyzing treatment data (i.e. corpus use). In so doing, we found that five studies could be divided into multiple unique samples because they had separable sub-populations (Stevens 1991; Cobb 1999; Tongpoon 2009; Rezaee *et al.* 2015; Vyatkina 2016). For example, Stevens (1991) had two separable treatment and control comparisons, so we could harvest two unique samples from the study. Furthermore, in most of the studies, the unique samples included multiple ESs depending on the type of measurements (see Supplementary material Appendix 5 for unique samples that have multiple ESs for measurement types).

In identifying and finalizing the moderator variables and their values, we had to be selective to ensure enough statistical power for regression analyses of moderator variables. For features that may have a bearing on the effectiveness of corpus use, we identified 10 variables in the

following two categories: (i) publication and population data and (ii) treatment data. Table 1 shows the descriptive statistics of the identified variables and their values based on our coding scheme.

Table 1

Descriptive statistics of independent variables

Variables	Posttest effect sizes			Follow-up effect sizes		
	Proportion of each value (number of ES/total number of ES) (per cent)	**Number of ESs (n = 77)**	**Number of unique samples (k = 38)**	**Proportion of each value (number of ES/total number of ES) (per cent)**	**Number of ESs (n = 34)**	**Number of unique samples (k = 13)**
1. Publication and population data						
(1) Publication date	Mean = 2,009.55	Minimum = 1,991	Maximum = 2,016.4	Mean = 2,011.94	Minimum = 2,005.9	Maximum = 2,015.8
(2) Publication type						
A. Journal article	73	56	30	47	16	8
B. PhD dissertation	23	18	5	53	18	5
C. Conference paper/ Book chapter	4	3	3	0	0	0
(3) Region						
A. Asia	43	33	14	65	22	7
B. Middle East	29	22	15	24	8	5
C. Other	29	22	9	12	4	1
(4) L2 proficiency						
A. Low	26	20	9	29	10	3
B. Intermediate	61	47	22	59	20	9
C. High	6	5	2	12	4	1
D. Mixed	6	5	5	0	0	0
(5) Specialty						
A. Languages	17	13	8	21	7	3
B. Other	25	19	11	3	1	1
C. Mixed	58	45	19	76	26	9
2. Treatment data						
(1) Interaction type						
A. Paper-based	30	23	13	15	5	3
B. CALL program	16	12	7	0	0	0
C. Concordancer	49	38	15	76	26	8
D. Mixed	5	4	3	9	3	2
(2) Corpus type						
A. Public corpus (e.g. COCA, BNC, Brown, OANC)	62	48	22	91	31	11
B. Local corpus (e.g. own, specialized, graded)	19	15	9	9	3	2
C. Pre-selected concordance lines (e.g. corpus-informed materials)	18	14	7	0	0	0
(3) L2 vocabulary dimension*						

Variables	Posttest effect sizes			Follow-up effect sizes		
	Proportion of each value (number of ES/total number of ES) (per cent)	Number of ESs (n = 77)	Number of unique samples (k = 38)	Proportion of each value (number of ES/total number of ES) (per cent)	Number of ESs (n = 34)	Number of unique samples (k = 13)
A. Precise knowledge	29	22	21	24	8	8
B. In-depth knowledge	53	41	32	50	17	12
C. Productive use ability	18	14	10	26	9	5
(4) Training						
A. Not received	16	12	7	0	0	0
B. Received	84	65	31	100	34	13
(5) Duration						
A. Short (<2 h in total or only 1 session)	25	19	8	12	4	1
B. Medium (about 3–9 sessions)	43	33	14	59	20	7
C. Long (≥10 sessions in total)	32	25	16	29	10	5

*The total number of unique samples for this variable is higher than 38 and 13 for posttest and follow-up effect sizes, respectively, because a unique sample may have multiple effect sizes to measure different dimensions of L2 vocabulary knowledge.

Publication and population data

Above all, for the publication data, two variables were coded. First, the date of publication (month and year) was coded. Second, the types of publication were coded (journal article, PhD dissertation, and conference paper or book chapter). This variable shows whether a study was peer-reviewed for the publication, in accordance with the types of publication. We found that most of the posttest ESs came from journal articles (56 posttest ESs, 73 per cent); however, PhD dissertations had a higher proportion of follow-up ESs (18 follow-up ESs, 53 per cent) than did other publication types.

For the population data, three variables were chosen. First, participants' countries of origin were checked. On the initial assumption that the participants have different L1s across studies, we believed that including this difference in the equations would contribute to more reliable results. As the second variable, and for the same reason, learners' overall L2 proficiency was coded. For example, any reported levels of L2 proficiency can be noted, as found in the following official tests, which were frequently available:

- Test of English as a Foreign Language (TOEFL) and Test of English for International Communication (TOEIC) developed by Educational Testing Service (ETS)
- International English Language Testing System (IELTS) and Cambridge English: Preliminary (PET) developed by Cambridge English Language Assessment.

Perhaps because of the aforementioned barriers of DDL, we found rather few ESs from studies of learners with low L2 proficiency (20 posttest ESs, 26 per cent; 10 follow-up ESs, 29 per cent).

For the third variable, on the initial assumption that students specializing in L2 learning would excel in corpus use, we checked the students' specialties, which included those related to language (e.g. English majors), other specialties unrelated to language (e.g. Engineering, Medicine, Social Science), or mixed specialties (studies that included both language and non-language majors).

Treatment data

Five variables were coded for treatment data (i.e. corpus use) to include different ways of implementing corpus use in various contexts. First, assuming that using new technology (e.g. concordancing software) affects learners' experience with corpus materials (Boulton 2010), we checked how students interacted with corpus data (i.e. interaction type), including paper-based activity, CALL program (e.g. Sketch Engine; Kilgarriff *et al.* 2014), concordancer-based activity, or a combination thereof (e.g. paper-based + concordancer-based activities). We found that a majority of ESs came from studies that provided computer-based, hands-on concordancing activity (38 posttest ESs, 49 per cent; 26 follow-up ESs, 76 per cent).

Second, different types of corpus data were noted as well. Although most ESs came from studies that used pre-established public corpora (48 posttest ESs, 62 per cent; 31 follow-up ESs, 91 per cent), such as COCA, BNC, Brown, and OANC, we could identify situations where researchers created materials using other collections of language data, such as local corpora of authentic texts, students' textbooks, and graded readers, or they opted to provide concordance lines that were selected to be more suitable for learners (Frankenberg-Garcia 2012, 2014; Lee *et al.* 2017).

Third, different dimensions of L2 vocabulary knowledge were checked, and we coded the aforementioned three values: precise knowledge, in-depth knowledge, and productive use ability. In essence, how a study measures participants' learning outcomes largely determines the target dimension of L2 vocabulary knowledge. Precise knowledge is concerned with recall or recognition of the meaning of a target lexical item, which is arguably the most frequently investigated dimension in L2 vocabulary learning research. In-depth knowledge, on the other hand, is measured by investigating participants' knowledge about syntagmatic (e.g. collocation) and paradigmatic (e.g. synonym) relationships between a target lexical item and others. Productive use ability is measured by asking participants to produce sentences

containing a target lexical item. We found that about half of ESs came from studies that investigated the in-depth dimension of L2 vocabulary knowledge (41 posttest ESs, 53 per cent; 17 follow-up ESs, 50 per cent).

Fourth, whether participants were trained in corpus use prior to the vocabulary treatment was coded. Given that most of the collected studies focused on the extensive use of concordancer and concordance lines, most ESs came from studies that included a training opportunity (65 posttest ESs, 84 per cent; 34 follow-up ESs, 100 per cent).

Last, the duration of an intervention was taken into consideration. To generate comparable criteria, we decided to endorse Boulton and Cobb's (2017) criterion to measure the duration of the treatment: short (less than 2 h in total or only 1 session), medium (about 3–9 sessions), and long (10 sessions or more).

To ensure that the data set was generated reliably and accurately, a second rater was trained in the protocol based on the created rubric for the coding procedure. He passed a reliability test that consisted of all codes from eight selected studies (30 per cent of the total number of studies) by achieving 100 per cent agreement with the first rater. After the completion of the entire coding procedure, we found that the inter-coder agreement was perfect. See Supplementary material Appendix 5 for the structure and the full coding scheme of our data set.

Data analysis plan

To answer the research questions about the overall mean ES as well as the impacts of moderator variables, we used STATA 14 (with *meglm* command) to conduct a multilevel regression analysis with the posttest/follow-up ESs as the dependent variables. All the results of multilevel regression analyses were replicated in HLM 7 to confirm their robustness. We used a multilevel model because the unit of our data analysis is each ES (Level 1), but they are nested in each unique sample (Level 2); this hierarchy of the data structure does not satisfy the independence assumption of conventional regression analysis (e.g. ordinary least squares regression). In other words, a problem comes up when all the observations are simply pooled in the conventional analysis without taking into consideration the nested design because the ESs nested in the same study would not be independent of each other (Raudenbush and Bryk 2002).

In multilevel models, on the other hand, we can avoid this issue because the analysis retains the nested design of each observation by distinguishing the Level 1 variance (i.e. the variance between the ESs within a unique sample) and the Level 2 variance (i.e. the variance between the unique samples; see Supplementary material Appendix 6 for an example equation of

multilevel model regression analysis). In this way, multilevel model analysis has one regression line for each unique sample, and they are parallel to the overall regression line (see Woltman *et al.* 2012 for the details of multilevel modeling). Furthermore, using multilevel regression analysis with the Level 1 units allowed us not only to compare and contrast these mean estimates to further analyze whether they are statistically different from each other but also to compute the adjusted mean ESs (i.e. predictive margins) for each value of the moderators after controlling for other moderators (or holding others constant).

Based on the discussion above, we first ran unconditional (i.e. no covariate) multilevel models to compute the mean ESs for posttests as well as follow-up tests. Our multilevel models differ from common multilevel modeling because our model is a *variance-known* model. In other words, we already know the Level 1 variance because our dependent variable consists of the computed ESs, and we know their sampling errors: the variance (the square of the standard errors) of the ESs (see Supplementary material Appendix 6 for the equations of multilevel regression analyses). Following Hox *et al.*'s (2010) theoretical suggestion and Pratt *et al.*'s (2016) practical guideline, we built a variance-known multilevel model by including the variance of the ESs as the Level 1 variance component. In this way, we could assign different weights to each unique sample according to the precision of the ESs within unique samples, given that the Level 1 variance component is used to estimate the Level 2 variance (Borenstein *et al.* 2009).

In addition, for the moderator analysis, we ran two more regression analyses with the variables of interest included on the right-hand side of the equations, one for the publication and population data and the other for the treatment data (see Supplementary material Appendix 6 for the equations). In doing so, we abided by the following two rules of thumb for regression analyses to ensure the robustness. First, as mentioned above, we ran two separate regression models for the moderator estimation (i.e. one for publication and population data, the other for treatment data) to have each model include a maximum of seven (number of posttest ESs/10 = 7.7) and three (number of follow-up ESs/10 = 3.4) predictors in the equations for the posttest ESs (n = 77) and the follow-up ESs (n = 34), respectively. This is to follow the one-in-ten rule —one of the most conservative perspectives to avoid risk of lack of degree of freedom and thus overfitting the model (see Vittinghoff and McCulloch 2007), which would mislead and bias inferential statistics of the model. For the follow-up ESs, however, we did not choose certain variables, but found that the variables excluded in the analyses could not be included in the model because the variables had either a multicollinearity issue or small samples.[3]

Second, to confirm the findings of multilevel regression analyses, we conducted a clustered regression analysis with the Huber–White-corrected estimator of variance and further compared its results with those of multilevel regression analyses (see Supplementary material

Appendix 7). Though this approach does not fully incorporate the hierarchical (i.e. multilevel) data structure, it produces the best linear unbiased estimates that are robust to heteroscedasticity; thus, this approach yields results similar to those of multilevel models (see Angeles and Mroz 2001; Wooldridge 2010). Overall, the results of this approach showed that multilevel regression analyses provide similar estimates (i.e. similar coefficients, standard errors, and *p* values), confirming the robustness of our findings.

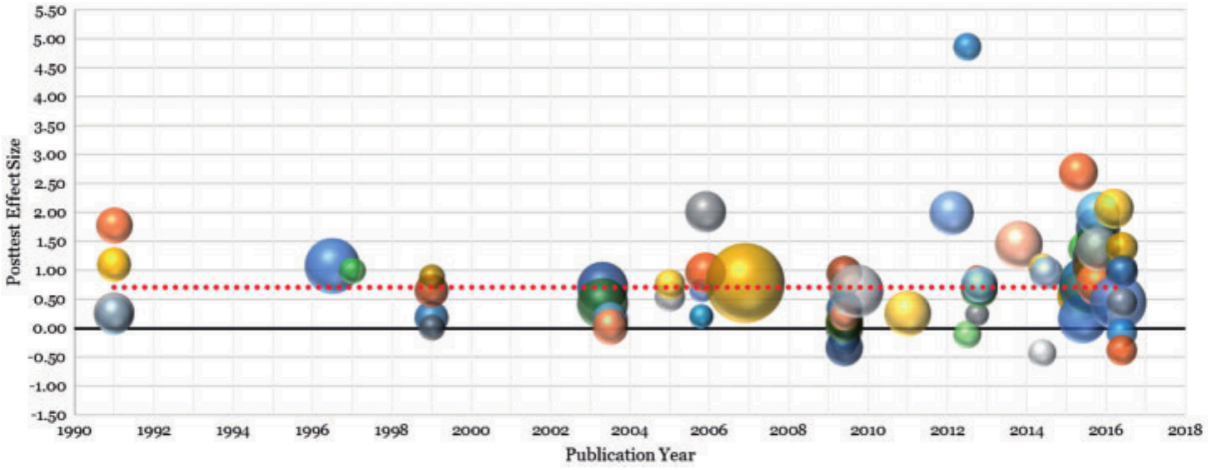


Figure 3: *Posttest effect sizes by year of publication. Bubbles represent each posttest ES. Bubble size is proportionate to the precision of each posttest ES; an ES with higher precision is indicated by a larger bubble*

RESULTS

An overview of the posttest ESs by publication year is given in Figure 3, which includes bubbles for each posttest ES. The bubbles' size is proportionate to the precision of each posttest ES; therefore, an ES with a high precision (i.e. low standard error) or a large sample size has a bigger bubble. This figure reveals that the studies included began in the 1990s and that a relatively large concentration emerged in the 2010s. In particular, about 60 per cent of the included studies (i.e. 17 studies of 29) were conducted in the 2010s, perhaps because the broader technical access, mentioned in our Introduction, facilitated dissemination and use of hands-on corpus-related programs in a wide range of pedagogical contexts (Godwin-Jones 2001; Reppen 2010; Lee and Lee 2015). Nevertheless, we found no association between the magnitude of posttest ESs and publication date ($b = 0.01$, $SE = 0.01$, $p = .24$).

Research Question 1: How effective is corpus use in improving L2 vocabulary learning?

Table 2 presents the weighted mean ESs as well as the variance components as the mean ES estimates. For the posttest ESs, we found a medium-sized effect of corpus use on L2 vocabulary learning ($g = 0.74$, $SE = 0.09$, $p < .001$; 77 posttest ESs coming from 38 unique

samples), indicating that its impact is in the top 50 per cent of L2 learning tools and instruction in the field of applied linguistics based on the benchmark suggested by Plonsky and Oswald (2014). The variance components of this estimate indicated that, among the model's total variation, about 60 per cent came from within unique samples (i.e. $\sigma_{level\ 1}^2 = 0.18$, $SE = 0.07$, $p < .001$) and the remaining 40 per cent came from between unique samples (i.e. $\sigma_{level\ 2}^2 = 0.12$, $SE = 0.07$, $p < .001$). For the follow-up ESs, we found that the positive medium-sized effect of corpus use was long-term ($g = 0.64$, $SE = 0.17$, $p < .001$; 34 follow-up ESs coming from 13 unique samples). Furthermore, having both the posttest and follow-up ESs the same size (i.e. medium size; $g > 0.6$) is noteworthy, indicating that corpus use as a learning tool is effective in enhancing L2 vocabulary long-term retention (see Min 2008 for a review on L2 vocabulary learning and retention). For the variance components of the follow-up ES estimates, about 37 per cent of the model's total variation came from Level 1 and 63 per cent from Level 2. As a result, the findings confirmed that corpus use works as an active student-centered L2 vocabulary learning approach. Furthermore, the estimates of the variance components of the models revealed that there was significant variation in the ES estimates both within unique samples (Level 1) and between unique samples (Level 2), thus suggesting that the use of multilevel modeling was necessary.

Table 2
Mean ES estimates

ES estimates	Posttest ESs	Follow-up ESs
1. Weighted mean effect size	0.74*** (0.09), 95 per cent CI [0.57–0.91]	0.64*** (0.17), 95 per cent CI [0.31–0.97]
2. Variance components		
Level 1: ES level	0.18*** (0.07)	0.15*** (0.08)
Level 2: Study level	0.12*** (0.07)	0.26*** (0.14)
Number of ESs (*n*)	77	34
Number of unique samples (*k*)	38	13

Notes: CI=confidence interval.
***$p < .001$.

In addition to the multilevel ESs, we computed a single average ES per study and confirmed both the short-term and the long-term medium-sized effect of corpus use (38 posttest ESs for unique samples, $g = 0.78$, $SE = 0.08$, $p < .001$; 13 follow-up ESs for unique samples, $g = 0.70$, $SE = 0.15$, $p < .001$). Also, we present a forest plot of these average ESs across the studies in accordance with the standard meta-analysis practice in the field (see Supplementary material Appendix 2 for the equations and procedures of the ES calculation at study level, and see Supplementary material Appendix 8 for the ES estimates and the forest plots).

Research Question 2: What are the moderators that influence the magnitude of the effectiveness of corpus use?

Tables 3 and 4 describe the results of both simple and multiple regression analyses for the identified nine moderators from the publication, population, and treatment data. In particular, the columns for the simple regression analyses show the descriptive statistics for each moderator, such as numbers of ESs and unique samples, and weighted means for each value of the moderator. The columns of multiple regression analyses present the inferential statistics (i.e. the adjusted means and their contrasts) to describe how each value of the moderator influences the magnitude of the effectiveness of corpus use along with the other moderators.

Publication and population data

In Table 3, we have the descriptive and inferential statistics of the four variables in the publication and population data, including (i) publication type, (ii) region, (iii) L2 proficiency, and (iv) specialty. First, for the publication type, the simple regression for the posttests showed a medium effect for journal articles and a large effect for other publication types, such as conference papers and book chapters. However, there was a negligible marginal effect for PhD dissertations. Also, the follow-up ESs had a similar pattern, where we found a large effect for journal articles and a negligible effect for PhD dissertations. After controlling for other variables in the publication and population data (or keeping other variables at their averages), however, multiple regression analyses showed that there were no statistically significant differences for the posttest ESs, indicating that the publication types do not have different effect sizes of corpus use on the assumption that other publication and population variables are the same across the publication types. Nevertheless, for the follow-up ESs, we found a large, statistically significant difference between PhD dissertations and journal articles (*ES difference*; $g = 0.91$, $SE = 0.23$, $p < .001$). Similar findings were reported in one of the previous meta-analysis studies (Boulton and Cobb 2017), and we agree with their interpretation that among many possible reasons this finding may indicate both submission bias (i.e. researchers are more likely to submit studies with higher impact to a journal) and acceptance bias (i.e. journals are more likely to accept studies with higher impact). A similar issue was discussed earlier in the publication bias section.

Second, the posttest ESs of the region variable indicated that the L2 learners from the Middle East had a mean ES of 1.06 ($SE = 0.13$, $p < .001$), which is higher than the ESs of the learners in either Asia ($g = 0.49$, $SE = 0.12$, $p < .001$) or Europe and the USA ($g = 0.68$, $SE = 0.15$, $p < .001$). The adjusted means and their contrasts from multiple regression analyses showed that the higher performance of the Middle Eastern learners is statistically significant when compared to their peers (*ES difference*; $g = 0.52$, $SE = 0.20$, $p < .01$). This finding corresponds

to that of Boulton and Cobb (2017), who found this result to be counterintuitive given the similar deductive-oriented L2 learning cultures both in Asia and the Middle East. Moreover, they suggested that the lower ESs for Europe and the USA may be due to the similarity between DDL and traditional teaching in their regions.

Table 3

Descriptive and inferential statistics of moderator variables I (publication and population data)

Moderator variables	Posttest effect sizes					Follow-up effect sizes				
	Simple regression			Multiple regression		Simple regression			Multiple regression	
	n	***k***	Weighted means	Adjusted means	Contrasts	***n***	***k***	Weighted means	Adjusted means	Contrasts
1. Publication and population data										
(1) Publication type										
A. Journal article	56	30	0.79*** (0.09)	0.74*** (0.08)	vs. B: 0.31 (0.21)	16	8	1.01*** (0.14)	0.99*** (0.14)	–
B. PhD dissertation	18	5	0.29 (0.17)	0.42* (0.18)	vs. C: −0.66 (0.48)	18	5	0.06 (0.16)	0.08 (0.16)	vs. B: −0.91*** (0.23)
C. Other[a]	3	3	1.29*** (0.34)	1.08* (0.45)	vs. A: 0.34 (0.46)	–	–	–	–	–
(2) Region										
A. Asia	33	14	0.49*** (0.12)	0.53*** (0.11)	vs. B: −0.52** (0.19)	22	7	0.26 (0.16)	Omitted[d]	
B. Middle East	22	15	1.06*** (0.13)	1.05*** (0.13)	vs. C: 0.52** (0.20)	8	5	1.09*** (0.22)		
C. Other[b]	22	9	0.68*** (0.15)	0.53*** (0.15)	vs. A: −0.00 (0.20)	4	1	1.13** (0.38)		
(3) L2 proficiency										
A. Low	20	9	0.50** (0.16)	0.47*** (0.13)	vs. B: −0.23 (0.15)	10	3	0.18 (0.29)	0.29 (0.18)	vs. B: −0.28 (0.23)
B. Intermediate	47	22	0.71*** (0.10)	0.69*** (0.09)	vs. D: −0.05 (0.38)	20	9	0.73*** (0.18)	0.57*** (0.13)	vs. C: −0.23 (0.40)
C. High	5	2	1.11*** (0.32)	1.27*** (0.31)	vs. A: 0.80* (0.36)	4	1	1.13* (0.48)	0.79*(0.37)	vs. A: 0.51 (0.45)
D. Mixed	5	5	1.21*** (0.26)	0.74* (0.35)	vs. C: −0.52 (0.46)	–	–	–	–	–
(4) Specialty										
A. Languages	13	8	0.90*** (0.19)	0.62** (0.20)	vs. B: 0.08 (0.24)	7	3	1.03** (0.33)	0.36 (0.29)	vs. B: 0.13 (0.56)
B. Other[c]	19	11	0.81*** (0.17)	0.54*** (0.14)	vs. C: −0.22 (0.18)	1	1	0.77 (0.65)	0.23 (0.49)	vs. C: −0.33 (0.51)
C. Mixed	45	19	0.64*** (0.12)	0.75*** (0.09)	vs. A: 0.14 (0.24)	26	9	0.50** (0.18)	0.56*** (0.12)	vs. A: 0.20 (0.34)
Number of ESs (*n*)	77					34				
Number of unique samples (*k*)	38					13				

Notes: Standard errors are in parentheses. A dash in a cell indicates that the variable does not have relevance to the cell.

[a]For example, conference paper, book chapter.

[b]For example, Europe, the USA.

[c]For example, Engineering, Medical, Social Science.

[d]The value of Other in the variable and the value of High in the L2 proficiency variable in the follow-up ES section have identical observations (i.e. n =4, k =1), and this may cause a multicollinearity issue, so we omitted the variable in the equation to avoid this.

*p < .05; **p < .01; ***p < .001.

Table 4

Descriptive and inferential statistics of moderator variables II (treatment data)

Moderator variables	Posttest effect sizes					Follow-up effect sizes				
	Simple regression			Multiple regression		Simple regression			Multiple regression	
	n	*k*	Weighted means	Adjusted means	Contrasts	*n*	*k*	Weighted means	Adjusted means	Contrasts
2. Treatment data										
(1) Interaction type										
A. Paper-based	23	13	0.67*** (0.15)	0.55*** (0.14)	vs. D: −0.75* (0.34)	5	3	0.89** (0.34)	0.70*** (0.19)	vs. C: 0.31 (0.22)
B. CALL program	12	7	0.73*** (0.21)	0.70** (0.23)	vs. A: 0.15 (0.26)	–	–	–	–	–
C. Concordancer	38	15	0.71*** (0.13)	0.72*** (0.14)	vs. B: 0.02 (0.31)	26	8	0.46* (0.19)	0.39*** (0.09)	vs. D: −0.72* (0.29)
D. Mixed	4	3	1.24*** (0.32)	1.30*** (0.32)	vs. C: 0.58 (0.38)	3	2	1.06* (0.43)	1.11*** (0.27)	vs. A: −0.31 (0.22)
(2) Corpus type										
A. Public corpus[a]	48	22	0.72*** (0.11)	0.65*** (0.11)	vs. B: 0.05 (0.25)	31	11	0.57** (0.17)	Omitted[f]	
B. Local corpus[b]	15	9	0.77*** (0.19)	0.59** (0.20)	vs. C: −0.39 (0.36)	3	2	1.06* (0.44)		
C. Pre-selected[c]	14	7	0.78*** (0.21)	0.98*** (0.28)	vs. A: 0.34 (0.34)	–	–	–		
(3) L2 vocabulary dimension[d]										
A. Precise knowledge	22	21	0.46*** (0.13)	0.40** (0.13)	vs. B: −0.51** (0.16)	8	8	0.38 (0.20)	0.29 (0.17)	vs. B: −0.48* (0.23)
B. In-depth knowledge	41	32	0.92*** (0.10)	0.91*** (0.10)	vs. C: 0.36 (0.20)	17	12	0.88*** (0.16)	0.77*** (0.13)	vs. C: 0.59* (0.23)
C. Productive use ability	14	10	0.53** (0.19)	0.55** (0.18)	vs. A: 0.15 (0.21)	9	5	0.20 (0.24)	0.18 (0.17)	vs. A: −0.11 (0.24)
(4) Training										
A. Not received	12	7	0.76*** (0.21)	0.58** (0.22)		–	–		Omitted[g]	
B. Received	65	31	0.73*** (0.10)	0.72*** (0.08)	vs. A: 0.14 (0.24)	34	13	0.64*** (0.17)		
(5) Duration[e]										
A. Short	19	8	0.71*** (0.17)	0.68** (0.21)	vs. B: 0.13 (0.28)	4	1	1.13*** (0.27)	1.24*** (0.22)	vs. B: 1.01*** (0.25)
B. Medium	33	14	0.51*** (0.13)	0.55*** (0.13)	vs. C: −0.35 (0.18)	20	7	0.10 (0.12)	0.23* (0.11)	vs. C: −0.52* (0.21)
C. Long	25	16	0.97*** (0.13)	0.90*** (0.14)	vs. A: 0.22 (0.29)	10	5	1.08*** (0.16)	0.75*** (0.16)	vs. A: −0.49 (0.28)
Number of ESs (*n*)	77					34				
Number of unique samples (*k*)	38					13				

Note. Standard errors are in parentheses. A dash in a cell indicates that the variable does not have relevance to the cell.

[a]For example, COCA, BNC, Brown, OANC.

[b]For example, own, specialized, graded.

[c]For example, providing carefully selected or syntactically modified concordance lines; corpus-informed materials.

[d]The total number of unique samples for this variable is higher than 38 and 13 for posttest and follow-up effect sizes, respectively, because a unique sample may have multiple effect sizes to measure different dimensions of L2 vocabulary knowledge.

[e]Short: less than 2 h in total or only 1 session; Medium: about 3–9 sessions; and Long: 10 sessions or more in total.

[f]The value of Local corpus in the variable and the value of Mixed in the Interaction type variable in the follow-up ES section have identical observations (i.e. n =3, k =2), and this may cause a multicollinearity issue, so we omitted the variable in the equation to avoid this.

[g]The variable has only one value and does not have any reference category; therefore, we omitted the variable in the equation.

[*]$p < .05$, [**]$p < .01$, [***]$p < .001$.

Third, the L2 proficiency variable showed that, for the posttest ESs, corpus use had a small effect on learners with low L2 proficiency (g = 0.50, SE = 0.16, p < .01), but had either medium or large effects for those with intermediate (g = 0.71, SE = 0.10, p < .001) or high L2 proficiency (g = 1.11, SE = 0.32, p < .001). When holding other variables constant, we found a medium, statistically significant difference between the learners with low proficiency and those with high proficiency (*ES difference;* g = 0.80, SE = 0.36, p < .05). For the groups of learners with mixed L2 proficiency levels, we found that multiple regression analyses substantially lowered their adjusted means to 0.74 (SE = 0.35, p < .05). Although the simple regression indicated a large effect for this group of learners (g = 1.21, SE = 0.26, p < .001), we cautiously suggest (because it is based on very limited data; there were only five ESs coming from five unique samples) that by including covariates in the equation, multiple regression analyses produced more reliable results than the simple regression. For the follow-up ESs, we found that the small effect of corpus use became negligible in the long term for low proficiency levels (g = 0.18, SE = 0.29, p > .05), but positive effects on L2 vocabulary learning for the learners with intermediate and high L2 proficiency remained (although from very limited data, as there were only four ESs coming from one unique samples for high proficiency levels). Taken together, we found that corpus use could be more effective for intermediate and high proficiency learners than for low proficiency learners.

Fourth, for the specialty variable, we found that the learners performed well in their corpus-based instruction/activities no matter what their specialties were. The simple regressions for both the posttest and follow-up ESs show that students specializing in the language-related disciplines tended to have slightly higher posttest (g = 0.90, SE = 0.19, p < .001) as well as follow-up ESs (g = 1.03, SE = 0.33, p < .01); however, after keeping other variables at their averages, we could not find any statistically significant difference between the specialties. This finding did not agree with our initial assumption, but it could be explained by the dominant role of the L2 proficiency level variable included in multiple regression analyses. In other words, we can assume that learners' specialty does not have a large effect on L2 vocabulary learning via corpus use once we control for L2 proficiency.

Treatment data

In Table 4, we present the descriptive and inferential statistics of the five variables for the treatment data, including (i) interaction type, (ii) corpus type, (iii) L2 vocabulary, (iv) training, and (v) duration. First, for the interaction type, we found that paper-based, CALL program-based, and concordancer-based activities or instruction all had a medium effect on improving L2 vocabulary knowledge (g = 0.67, SE = 0.15, p < .001; g = 0.73, SE = 0.21, p < .001; g = 0.71, SE = 0.13, p < .001, respectively), and that using a combination of these types

(e.g. paper-based activity + concordancer) had the largest ES (g = 1.24, SE = 0.32, p < .001). Multiple regression analyses for the posttest ESs revealed that there was a medium-sized difference between using only paper-based and using paper-based + additional interaction types (*ES difference*; g = 0.75, SE = 0.34, p < .05). Similarly, multiple regression analyses for the follow-up ESs revealed that using multiple interaction types for corpus use produced superior results in the long term as well. Although the analyses of the follow-up ESs appeared to favor the paper-based type of corpus use over the concordancer-based type, the ES difference was not significant (*ES difference;* g = 0.31, SE = 0.22, p > .05).

Second, for the posttest ESs for the second moderator in the treatment data, we found medium-sized effects of corpus use on L2 vocabulary learning for all the different corpus types. According to the adjusted means from multiple regression analyses, we found that careful selection of concordance lines by teachers or researchers had a large impact on improving L2 vocabulary knowledge (g = 0.98, SE = 0.28, p < .001), whereas using either public or local corpora had a medium-sized ES (g = 0.65, SE = 0.11, p < .001; g = 0.59, SE = 0.20, p < .01; respectively). For the follow-up ESs, the positive effects of using either public or local corpora remained long-term. In regard to the remaining coefficients, their small sample sizes did not allow us to explain further.

Third, the analyses of the posttest ESs for the L2 vocabulary variable indicated that corpus use improved all three dimensions of L2 vocabulary knowledge. In particular, although corpus use had a large effect on improving in-depth knowledge (g = 0.92, SE = 0.10, p < .001), it had only a small effect for precise knowledge and productive use ability (g = 0.46, SE = 0.13, p < .001; g = 0.53, SE = 0.19, p < .01; respectively). For the ES difference between the dimensions, the results of multiple regression analyses revealed that corpus use is more effective for expanding in-depth knowledge of L2 vocabulary than for increasing learners' precise knowledge of L2 vocabulary (*ES difference*; g = 0.51, SE = 0.16, p < .01). For the follow-up ESs, corpus use was effective only for in-depth knowledge (g = 0.88, SE = 0.16, p < .001), as its effects on improving the other dimensions became negligible in the long term. In brief, this pattern corresponds to the findings of a recent meta-analysis on computer-mediated textual glosses (Abraham 2008), where learners benefited from glosses less in terms of productive lexical knowledge than in terms of receptive knowledge (i.e. precise + in-depth knowledge). It also accords with the general structure of vocabulary knowledge among L2 learners, with receptive knowledge being greater than productive knowledge (Laufer and Paribakht 1998). It is interesting to note that corpora use led to greater gains in in-depth knowledge than in precise (i.e. definitional) knowledge, and a possible explanation for this

finding may be related to the nature of corpus use that offers multiple samples of target vocabulary. This finding is one of the unique contributions of the present study and we include an in-depth discussion in the following section.

Fourth, we found that the learners performed almost equally well for their L2 vocabulary learning based on corpus use whether they had had a training opportunity. Although multiple regression analyses revealed that receiving training could have a slightly larger effect than not receiving any training when keeping other treatment-related variables at their averages, the ES difference was negligible and was not statistically significant (*ES difference*; $g = 0.14$, $SE = 0.24$, $p > .05$). All the studies that had follow-up ESs provided training opportunities, so no further investigation was conducted for this variable.

Fifth, for the duration of instruction, we found that the interventions with 10 sessions or more had larger ESs than interventions with less than 10 sessions. However, there was no statistically significant difference across the different lengths of the intervention, according to the results of multiple regression analyses. Further exploration of the follow-up ESs did not give us a clear picture of the impact of the duration of instruction, again because the samples were small.

DISCUSSION AND CONCLUSION

Our meta-analysis was conducted to investigate the overall effect of corpus use on L2 vocabulary learning and to identify moderators that may influence the effectiveness of corpus use. Based on the calculated 77 posttest ESs from 38 unique samples and 34 follow-up ESs from 13 unique samples, our meta-analysis showed a medium-sized effect on L2 vocabulary learning, with the greatest benefits for promoting in-depth knowledge to learners who have at least intermediate L2 proficiency. Corpus use was also more effective when the concordance lines were purposely selected and provided and when learning materials were given along with hands-on corpus-use opportunities. Moreover, we found that corpus use is still effective even without prior training and remains effective regardless of the corpus type or the length of the intervention. In the following, we discuss our findings in terms of corpus use for constructive L2 vocabulary learning and different dimensions of L2 vocabulary knowledge and also discuss evidence-based considerations in implementing corpus use.

Overview: Corpus use for constructive L2 vocabulary learning

Overall, our finding of an overall medium effect in multilevel meta-analyses of corpus use for both the short ($g = 0.74$) and the long term ($g = 0.64$) is an important indication of the self-driven construction of L2 vocabulary learning. Given that the studies included in our meta-analysis were mostly concerned with ‘direct’ (as opposed to ‘indirect’) use of corpora in

language teaching (Leech 1997), in which learners gain direct access to corpus materials, we assume that the learners in the sampled studies apparently explored the data independently for their linguistic inquiry and language learning. This process, also known as DDL (Johns 1991), has been at the heart of corpus use in language teaching (Leech 1997) and is associated with important interrelated educational constructs, such as learner autonomy (Gavioli 2009), motivation (Curado Fuentes 2015), and discovery learning/constructivism (Flowerdew 2015).

For this reason, the fact that learners construct or discover L2 vocabulary knowledge by using a corpus-based approach has important implications for language teaching. As Bernardini (2004) highlights, discovery learning 'encourages learners to follow their own interests' (p. 23), and learners may pave different learning paths even with the same learning materials. In other words, they independently and individually construct their own knowledge. Bernardini further notes that this change in the learning approach influences the previous roles of teacher and learners as well as those of material developers and curriculum designers. In view of constructivism, although the findings of this meta-analysis point to the value of a corpus-based approach, *how* learners construct their knowledge deserves more attention. As Flowerdew (2015) suggests, learners are expected to tackle corpus materials with 'higher order cognitive skills' (p. 18), which would interact with the learners' current L2 proficiency level and other moderators (see below for our discussion on this issue). This very complicated process would be better explored through a more qualitative lens; such an investigation, integrated with a quantitative approach, could help teachers and researchers contemplate more effective ways to develop individualized instruction.

Corpus use and dimensions of L2 vocabulary knowledge

Furthermore, our focus on specific dimensions of L2 vocabulary knowledge led us to understand that several moderators may come into play in the effectiveness of corpus use in L2 vocabulary learning. The most salient and intriguing finding from this approach is related to the dimensions of L2 vocabulary knowledge, with in-depth properties of target lexical items being learnt best. A possible explanation for this finding may lie in the KWIC format, which offers multiple contextual instances of vocabulary (Johns 1994). This feature helps learners refer to multiple samples of how a target lexical item is used differently in various contexts. Learners would then be able to understand in-depth properties of a target lexical item, such as paradigmatic and syntagmatic relationships between a target lexical item and others (Wolter 2006), which accords with predictions made by proponents of corpus use in vocabulary teaching and the DDL approach (Johns 1991; Kita and Ogata 1997; Cobb 1999; Hill 2000).

Moreover, this finding fits in with views of researchers, such as Bernardini (2004) or Flowerdew (2015), who introduced diverse ways of using DDL in L2 instruction. Although they do not specifically refer to the vocabulary knowledge framework like the one used in this study (i.e. Henriksen 1999), it can be inferred from their use of expressions, such as 'collocates' and 'synonyms and/or antonyms', that in-depth knowledge is expected to be developed by learners' exploration of concordance lines. This is a promising finding for corpus use as an L2 vocabulary teaching approach, considering that this dimension of knowledge is not frequently dealt with in instructed L2 vocabulary research presumably because the primary focus in this literature is on establishing the link between form and meaning of target vocabulary (Schmitt 2008). On this point, Schmitt (2008: 334) suggests that an explicit approach be used first to establish this 'form-meaning link', and 'exposure approach' could later be used to develop more 'contextual knowledge' (i.e. in-depth knowledge). Although he did not elaborate on the details of this approach, we suggest, based on our findings, that corpus use could be an appropriate mechanism for this.

Evidence-based considerations for corpus use in L2 vocabulary learning

Given the considerations about implementing corpus use discussed above, such as the suitability of language data and mastery of corpus consultation skills, the findings of our meta-analysis offer some evidence-based suggestions on how to effectively implement DDL in L2 vocabulary learning. Above all, one noticeable factor that influences the effectiveness of corpus use is whether the language data provided in corpus use were suitable for the learners. In the initial stages of adapting corpus materials to classroom contexts, the belief among teachers that low proficiency learners cannot handle corpus data well for learning appears to have impeded its wide application (Leńko-Szymańska and Boulton 2015). Our finding that learners with low L2 proficiency do not benefit as much from corpus use confirms this belief. As Chujo *et al.* (2015: 111) note, 'if DDL is to be considered for low proficiency learners, there is a need to rethink available corpora'.

The results of the moderator analysis also pointed to the importance of some methodological features that could contribute to learners' engagement with the given corpus materials for their independent discovery learning. For example, we found that providing L2 learners with pre-selected, comprehensible concordance lines appears more effective in supporting their corpus-based activities, which corresponds to previous empirical findings (Frankenberg-Garcia 2012, 2014; Lee *et al.* 2017). Also, this is in line with recommendations about the use of corpus materials that are more finely tuned to learners' L2 proficiency level (Allan 2009; Poole 2012; Chujo *et al.* 2015; Lee *et al.* 2017) and about learner-friendly concordancer software specifically designed for L2 learning (Chujo *et al.* 2015; Lee *et al.* 2015). We also verified that

both paper-based and computer-based corpus activities have medium-size effects on improving L2 vocabulary learning, thus suggesting that the way corpus materials are presented to learners may not be a critical variable. In addition, confirming Boulton's (2009) finding that DDL can be effective even for untrained learners, the results showed that corpus-based activity without any prior training is just as effective as when training was received. Also, our finding that the duration of corpus-based activity was not associated with the effectiveness of corpus use corresponds to previous findings (Boulton and Cobb 2017).

Taken together, we believe that learners' L2 proficiency plays the most significant role in DDL activities, though the evidence we found from the meta-analysis is based on limited data (there were only five ESs from two unique samples for high proficiency levels). Given that independent discovery learning requires substantial levels of linguistic knowledge and inference on the part of learners (Johns 1991), the importance of learners' L2 proficiency is expected. It is true, however, that some recent studies (Vyatkina 2016) revealed that DDL could also be effective for beginning learners, and the findings of our meta-analysis support this argument. Nevertheless, considering the different ESs for each L2 proficiency level—the small effect for beginners and medium or large effects for more advanced learners—we found that more proficient learners might benefit more extensively. For example, the ES difference between the low and high proficiency levels for the posttest ESs is around 0.80; that is, the mean of the group of students at the high L2 level is higher than about the 79th percentile of the group of students with low L2 proficiency (Cohen 1988). In other words, less than 21 per cent of low proficiency students who participated in the included studies could perform as well as or better than a student with high L2 proficiency—all other variables being equal. For this reason, we believe that the difference between the adjusted means of L2 proficiency levels has to be approached differently from the other estimated moderators. As Flowerdew (2015: 18) suggests, 'constructivism [which theoretically underlies the DDL approach] may not be ideal for all students' because of individual learner differences. We think that L2 proficiency may be an important factor among the differences that influence the effectiveness of corpus use for L2 vocabulary learning and deserves more attention in future research.

Limitations and suggestions

Despite the contributory findings, this study is not without its limitations. The sample size was not large enough to draw more reliable and comprehensive results, particularly for L2 proficiency. For example, although we could draw statistically significant estimates for this issue (i.e. weighted and adjusted means for each value of the variable), there were only five posttest ESs (6 per cent) and four follow-up ESs (12 per cent) for high L2 proficiency. Although the intermediate L2 proficiency level we used could, in theory, be divided into low

(B1) and high intermediate (B2) levels to match the Common European Framework of Reference for Languages (CEFR) guideline, making subjective judgments would not be methodologically justified because many of the included studies did not provide detailed information from which we could draw an objective distinction. Mainly because of the small sample size for this moderator, we did not have enough statistical power to compute interaction effects between moderator variables accurately. Similarly, in addition to the moderators identified in the included studies, there may be other moderators that could influence L2 vocabulary learning, such as corpus size, purpose of using the corpus (e.g. to answer specific linguistic inquiries or to acquire L2 vocabulary), or contexts (e.g. foreign or L2 learning). Unfortunately, our small sample size did not allow us to conduct a reliable moderator estimation for these variables. Having more studies on pedagogical use of a corpus would make it possible to perform such an analysis and, thus, possibly suggest further pedagogical implications.

In light of these limitations, which call for the accumulation of more empirical evidence, we recommend that future research in L2 studies includes clear descriptions of students' L2 proficiency levels as well as information about their L2 vocabulary teaching contexts, to minimize meta-analysts' 'informed guesswork' (Boulton and Cobb 2017: 28). In addition, an investigation into interactions between L2 proficiency level and the dimensions of L2 vocabulary knowledge should yield important pedagogical implications for effective corpus use for L2 vocabulary learning. Finally, further development of computer technology may continue to spawn more effective and efficient means of incorporating corpus use into L2 vocabulary learning, which will also, in turn, merit empirical investigation.

NOTES

1 We will follow Plonsky and Oswald's (2014) ES interpretation for L2 studies, where for between-group comparisons (i.e. control/experimental group comparisons) 0.4, 0.6, and 0.9 are considered small, medium, and large ESs, respectively, and 0.6, 1.0, and 1.4 for within-group comparisons (i.e. pre/posttest designs). Given that ESs should be interpreted within a specific field, Plonsky and Oswald analyzed 346 primary studies and 91 meta-analyses in L2 research and took the 25th, 50th, and 75th percentiles as indicators for the ES interpretation.

2 It should be noted here that having multiple ESs for a study does not contribute to the overall results of meta-analysis multiple times more than a study with only one ES. When using multilevel modeling, it explicitly recognizes the membership of ESs (i.e. the nested structure; ESs are nested in studies) and that ESs from the same study are more similar to

each other than ESs from other studies. In terms of different contributions of ES to the overall meta-analysis, the sample size (or the standard error) of each ES plays a significant role. Details are discussed in the data analysis section.

3 For example, the value of Other in the Region variable and the value of High in the L2 proficiency variable in the follow-up ES section have identical observations (see Table 3); this may cause a multicollinearity issue, so we omitted the variable in the equation to avoid this. In regard to the small sample issue, we can see that the Training variable in the follow-up ES section has only one value (see Table 4), and the variable does not have any reference category.

SUPPLEMENTARY DATA

Supplementary material is available at *Applied Linguistics* online.

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